Curriculum Vitae

Sung Eun Kim

- **Research Scientist**, Department of Biomedical Informatics, Harvard Medical School

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Professional Profile:

Board-certified orthopedic surgeon and clinician–scientist currently serving as a **Research Scientist in the Department of Biomedical Informatics at Harvard Medical School**. My research focuses on developing and evaluating artificial intelligence systems for medicine, spanning **multimodal medical imaging, clinical AI and large language models, longitudinal disease and outcome modeling, and autonomous clinical systems**. My work includes dynamic simulation frameworks for evaluating AI in realistic clinical environments, vision-language models for grounded reasoning across 2D and 3D medical imaging, clinically grounded evaluation of generative AI, and robotic systems for procedural and surgical assistance. Building on a clinical foundation in orthopedic surgery and extensive research in musculoskeletal imaging, osteoarthritis, and surgical outcomes, I aim to develop **reliable, clinically grounded AI systems that can reason across complex medical data and improve diagnosis, decision-making, forecasting, and patient care**.

Education:

Master of Science in Orthopedic Surgery
Seoul National University, College of Medicine, Seoul, Republic of Korea
February 2023

- Thesis: "Dedifferentiated Liposarcoma of the Extremities: Characteristics and Outcomes"
- Advisor: Prof. Ilkyu Han

Board-Certified Orthopedic Surgeon
Seoul National University Hospital, Seoul, Republic of Korea
March 2022

Doctor of Medicine (M.D.)
Seoul National University, College of Medicine, Seoul, Republic of Korea
February 2014

Professional Experience:

Research Scientist Oct 2026 – Present
Department of Biomedical Informatics, Harvard Medical School, Boston, USA

Postdoctoral Research Fellow Sep 2024 – Sep 2026
Department of Biomedical Informatics, Harvard Medical School, Boston, USA
Advisor: Pranav Rajpurkar, Ph.D.

Associate Research Professor Sep 2024 – Aug 2026
Biomedical Research Institute, Seoul National University Hospital, Seoul, Republic of Korea

Clinical Instructor (Knee surgery) Mar 2022 – Jul 2024
Department of Orthopedic Surgery, Seoul National University Hospital, Seoul, Republic of Korea
Advisors: Du Hyun Ro, M.D., Hyuk-Soo Han, M.D., Ph.D., and Myung Chul Lee, M.D., Ph. D.

Resident Mar 2018 – Feb 2022
Department of Orthopedic Surgery, Seoul National University Hospital, Seoul, Republic of Korea

Intern May 2017 – Feb 2018
Seoul National University Hospital, Seoul, Republic of Korea

Epidemic Intelligence Service Officer (Public Health Doctor) Apr 2014 – Apr 2017
Jeju National Quarantine Station, Korea Disease Control and Prevention Agency, Jeju-do, Republic of Korea

Publications:

1. Luo L*, **Kim SE***, Zhang X, Kernbach JM, Kenia R, Acosta JN, Nathanson L, Haimovich A, Rodman A, Goh E, Chen J, Shah N, Kim D, Zou J, Mahmood F, Kather JN, Lungren M, Karthikesalingam A, Natarajan V, Topol E, Rajpurkar P. A clinical environment simulator for dynamic AI evaluation. Nat Med (2026). https://doi.org/10.1038/s41591-026-04252-6 (*first author).

2. Kim TM, Luo L, **Kim SE**, Willauer AN, Rajpurkar P. The White Coat Effect: Specialist Personas Drive Harmful Diagnostic Drift More Than Patient Personas in Clinical LLMs. Pacific Symposium on Biocomputing (Accepted PSB 2027)

3. **Kim SE**, Lee MC, Ro DH. Five-Year Prospective Multicenter Outcomes of an Anatomically Designed Total Knee System: Survivorship, Function, and Patient Satisfaction in 1,017 Knees. Clinics in orthopedic surgery. Accepted.

4. Song JW, **Kim SE**, Ro DH. Different osteoarthritis progression rates within early-stage knee osteoarthritis: a retrospective cohort study using machine learning cluster analysis from the osteoarthritis initiative database. Knee Surg Relat Res. 2026;38:55. doi.org/10.1186/s43019-026-00355-z

5. Yuan GC, Zhang X, **Kim SE**, Luo L, Rajpurkar P. Asclepius: An Adaptive Harness for Long-Horizon Clinical Agents. EMNLP 2026

6. **Kim SE**, Zhang X, Wang X, Rajpurkar P. SurgeCount: A Protocol-Guided Composition Harness for Bimanual Surgical Sponge Counting. Machine Learning for Health (ML4H) 2026

7. Banerjee O, **Kim SE**, Willauer AN, Kernbach JM, Zhou CY, Alomaish AR, Alghamdi RAS, Alomaish H, Baharoon M, Zhang X, Acosta JN, Rajpurkar P. ReXInTheWild: A Unified Benchmark for Medical Photograph Understanding. Machine Learning for Health (ML4H) 2026, arXiv:2603.19517.

8. Wang X, Yang Z, **Kim SE**, Hardy R, Sampson MP, Reddy A, Pal A, Zhang X, Luo L, Rajpurkar P. SimRetract: Zero-Shot Sim-to-Real Robotic Assistance for Retractor Take-Over in Open Surgery. Machine Learning for Health (ML4H) 2026

9. Chen E, Banerjee O, Pal A, Wang X, Yang Z, Zhou CY, He K, Hardy R, Pancholy M, **Kim SE**, Willauer AN, Kim DA, Rajpurkar P. ECGDreamSuite: A Benchmark for ECG Waveform Generation in Patient World Models. Machine Learning for Health (ML4H) 2026

10. Rao VM*, Chen E*, Saenz A*, Banerjee O, Marklund H, Bijnens L, **Kim SE**, Zucker E, Shelmerdine S, Scala M, Niedermair J, Stranger N, Tschauner S, Schmölzer GM, Binzhi T, Yu L, Veloza Vega MJ, Osorio Valencia JS, Jaramillo Robledo O, López Uribe OJ, Afat S, Küstner T, Tayebi Arasteh S, Truhn D, Pomykala KL, Kleesiek J, Afifah NH, Irfan RM, Shahriarirad R, Erfani A, Amirav I, Be'er M, Mitsuyama Y, Ueda D, Abi-Ghanem AS, Al Zakleet S, Mohaed ASh, Mustapha WIBW, Nizami N, Khan R, Al-Kharazi K, Iqbal Z, Al Shahrani A, Kashgari A, Prodanovic T, Prodanovic N, Desmond LSW, Hallinan JTPD, Nongena P, Cooke ML,

Bret-Zurita M, Alonso-Ojembarrena A, Nemes A, Lin YC, Lin WT, Fang WH, Lin C, Mahayosnond A, Sriswasdi S, Khumrin P, Tanmuk S, Arman D, Cömert S, Alan S, Tandircioglu UA, Mong DA, Chung E, Sadler M, McBee MP, Jones RH, Rajpurkar P. Repurposing Adult AI Radiograph Models for Neonatal Care Through Continual Learning: An International Multicenter Study. Nature Biomedical Engineering. Accepted.

11. Wang M, Luo L, **Kim SE**, Cao F, Kim DA, Rajpurkar P. Open-Ended Clinical Text Generation for Acute Care: Applying Reinforcement Learning with Clinically Grounded Rewards Proceedings of the 7th Conference on Health, Inference, and Learning, PMLR 333:966-984, 2026.

12. Wang X, Yang Z, Zhang X, **Kim SE**, Hardy R, Rajpurkar P. Imitation Learning for Robot Assistance in Open Surgery: A Multi-Policy Evaluation on Suture Following. Machine Learning for Healthcare (MLHC 2026).

13. Baharoon M, Heintz T, Raissi S, Alabbad M, Alhammad M, AlOmaish H, **Kim SE**, Banerjee O, Rajpurkar P. CRIMSON: A Clinically-Grounded LLM-Based Metric for Generative Radiology Report Evaluation. Medical Image Computing and Computer Assisted Intervention (MICCAI 2026), early accept.

14. Wang X, Zhang X, **Kim SE**, Pal A, Rajpurkar P. ReXSonoVQA: A Video QA Benchmark for Procedure-Centric Ultrasound Understanding. Conference on Health, Inference, and Learning (CHIL 2026).

15. Baharoon M, Luo L, Moritz M, Kumar A, **Kim SE**, Zhang X, Zhu M, Alabbad MH, Alhazmi MS, Mistry NP, Bijnens L, Kleinschmidt KR, Chrisler B, Suryadevara S, Jaliparthi SSD, Prudlo NM, Marino MD, Palacio J, Akula R, Zhou D, Zhou HY, Hamamci IE, Adams SJ, Alomaish HR, Rajpurkar P. ReXGroundingCT: A 3D Chest CT Dataset for Segmentation of Findings from Free-Text Reports. NEJM AI. 2026;3(7). doi.org/10.1056/AIdbp2501220

16. Kim TM, Luo L, **Kim SE**, Manrai AK, Topol E, Rajpurkar P. The Doctor Will Agree With You Now: Sycophancy of Large Language Models in Multi-Turn Medical Conversations, HeaLing Workshop at the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026).

17. Yuan GC, Zhang X, **Kim SE**, Rajpurkar P. Do Mixed-Vendor Multi-Agent LLMs Improve Clinical Diagnosis?, HeaLing Workshop at the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026).

18. Park J, Persson A, Martin RK, Inderhaug E, **Kim SE**, Kim S, Kwak D, Ro DH. Effects of axial malrotation on posterior tibial slope measurement: a digitally reconstructed radiograph study enabling automated quality assessment. Knee Surg Relat Res. 2026;38:3. doi.org/10.1186/s43019-026-00303-x

19. Lee JY, Park JE, **Kim SE**, Choi BS, Lee MC, Chang CB, Han HS, Park HY, Ro DH. Preoperative low cognitive function and somatic psychological symptoms are associated with an increased risk of postoperative delirium following total knee arthroplasty: a prospective cohort study. J Bone Joint Surg Am. doi.org/10.2106/JBJS.25.00392

20. Kwak D*, **Kim SE***, Park J, Han H-S, Choi BS, Ro DH. Posterior tibial slope as a risk factor for anterior cruciate ligament tears: a retrospective study using automated measurement. **Arthrosc Orthop Sports Med**. 2025 Nov 1;12:64-69. doi: 10.14517/aosm25006 (*first author).

21. Sambara S*, **Kim SE***, Zhang X, Luo L, Johri S, Baharoon M, Ro DH, Rajpurkar P. 3DReasonKnee: Advancing Grounded Reasoning in Medical Vision Language Models. Pacific Symposium on Biocomputing (PSB) 31:99-113(2026). (*first author)

22. Baharoon M, Raissi S, Jun JS, Heintz T, Alabbad MH, Alburkani AM, **Kim SE**, Kleinschmidt K, Alhumaydhi AO, Alghamdi MMG, Palacio JF, Bukhaytan M, Prudlo NM, Akula R, Chrisler B, Galligos B, Almutairi MO,

Alanazi MM, Alrashdi NM, Hwang JJ, Jaliparthi SSD, Nelson LD, Nguyen N, Suryadevara S, Kim S, Mohammed MF, Semenov Y, Yu K-H, Aljouie A, Alomaish H, Rodman A, Rajpurkar P. RadGame: An AI-Powered Platform for Radiology Education. Machine Learning for Health (ML4H) 2025. doi.org/10.48550/arXiv.2509.13270

23. Acosta JN, Adams SJ, Kernbach JM, Hardy R, **Kim SE**, Luo L, Zhang X, Johri S, Baharoon M, Rajpurkar P. Voice-guided Orchestrated Intelligence for Clinical Evaluation (VOICE): A Voice AI Agent System for Prehospital Stroke Assessment. 2025 Second International Conference on Artificial Intelligence for Medicine, Health and Care (AIxMHC). 2025:14-21. doi.org/10.1109/AIxMHC65380.2025.00013

24. Baharoon M, Luo L, Moritz M, Kumar A, **Kim SE**, Zhang X, Zhu M, Kleinschmidt K, Jaliparthi SSD, Suryadevara S, Akula R, Marino M, Lei W, Hamamci IE, Rajpurkar P. State-of-the-art text-prompted medical segmentation models struggle to ground chest CT findings. Proceedings of the 10th Machine Learning for Healthcare (MLHC) Conference, in Proceedings of Machine Learning Research, Vol. 298, 2025. https://proceedings.mlr.press/v298/baharoon25a.html

25. Hardy R, **Kim SE**, Ro DH, Rajpurkar P. ReXTrust: A Model for Fine-Grained Hallucination Detection in AI-Generated Radiology Reports. Proceedings of The First AAAI Bridge Program on AI for Medicine and Healthcare, 2025, PMLR 281:173-182.

26. Heiman A, Zhang X, Chen E, **Kim SE**, Rajpurkar P. FactCheXcker: Mitigating Measurement Hallucinations in Chest X-ray Report Generation Models. Conference on Computer Vision and Pattern Recognition (CVPR) 2025, https://doi.org/10.48550/arXiv.2411.18672

27. Gimm G, Jeon B, **Kim SE**, Kim BS, Han HS, Kim S. A development of machine learning models to preoperatively predict insufficient clinical improvement after total knee arthroplasty. J Orthop Surg Res. 2025 Aug 20;20(1):778. doi: 10.1186/s13018-025-06206-z.

28. Park J, Chaar O, Narayanakurup J, Abdelhamead ASA, Ro DH, **Kim SE**. Do knee alignment patterns differ between Middle Eastern and East Asian populations? A propensity-matched analysis using artificial intelligence. Knee Surg Relat Res. 2025 Mar 3;37(1):11. doi: 10.1186/s43019-025-00261-w (corresponding author).

29. Ryu B, Nam JW, Ro DH, Martin RK, Inderhaug E, Persson A, Haaland S, **Kim SE**. Automated Posterior Tibial Slope Measurement Using Lateral Knee Radiographs: A Novel Landmark-Based Approach Using Deep Learning. Orthop J Sports Med. 2025 Apr 22;13(4):23259671251331067. doi: 10.1177/23259671251331067 (corresponding author).

30. Martin RK, Haaland S, Persson A, **Kim SE**, Ryu B, Nam JW, Ro DH, Inderhaug E. External validation of a novel landmark-based deep learning automated tibial slope measurement algorithm applied on short radiographs from ACL injured patients. Orthop J Sports Med. 2025 May 5;13(5):23259671251333607. doi: 10.1177/23259671251333607.

31. Zheng Z, Ryu, B, **Kim SE**, Song D, Kim SH, Park JW, Ro DH. Deep learning for automated hip fracture detection and classification: achieving superior accuracy. Bone Joint J. 2025 Feb 1;107-B(2):213-220. doi: 10.1302/0301-620X.107B2.BJJ-2024-0791.R1.

32. Kim JS, Choi BS, **Kim SE**, Lee YS, Lee DW, Ro DH. Machine learning-based prediction of contralateral knee osteoarthritis development using the Osteoarthritis Initiative and the Multicenter Osteoarthritis Study dataset. J Orthop Res. 2025 Mar;43(3):576-585. doi: 10.1002/jor.26018.

33. Park J, **Kim SE**, Kim B, Lee S, Lee JJ, and Ro DH. A deep learning based automatic two-dimensional digital templating model for total knee arthroplasty. Knee Surg Relat Res. 2024 Nov 27;36(1):38. doi:

10.1186/s43019-024-00240-7.

34. **Kim SE**, Kim SH, Lee JI, Han HS, Lee MC, Ro DH. Primary total knee arthroplasty for treating knees with neglected patellar dislocation. Medicina (Kaunas). 2024 Sep 13;60(9):1492. doi: 10.3390/medicina60091492.

35. Park JH, **Kim SE**, Kim W, Kim Y, Kim JH, Seo SW, Kim HS, Kim S, and Han I. Dedifferentiated Liposarcoma of the Extremities: A Korean Multi-Center Study of 107 Cases. BMC Cancer. 2024 Oct 10;24(1):1259. doi: 10.1186/s12885-024-13021-y.

36. **Kim SE**, Ro DH, Lee MC, Han HS. Can individual functional improvements be predicted in osteoarthritic patients after total knee arthroplasty? Knee Surg Relat Res. 2024 Oct 14;36(1):31. doi: 10.1186/s43019-024-00238-1.

37. **Kim SE**, Shin HS, Han HS. Superior Forgotten Joint Scores with Cruciate-Retaining Mobile bearings Versus Posterior-Stabilized Mobile and Fixed bearings in a contemporary Total Knee System. Knee. 2025 Mar:53:118-125. doi: 10.1016/j.knee.2024.11.003.

38. **Kim SE**, MacDessi S, Song D, Kim JI, Choi BS, Han HS, Ro DH. Coronal plane alignment of the knee (CPAK) type shifts towards constitutional varus with increasing Kellgren and Lawrence grade: a deep learning radiographic analysis of 17,365 knees. J Bone Joint Surg Am. 2024 Dec 04. doi: 10.2106/JBJS.24.00316.

39. Hussain I*, **Kim SE***, Kwon C, Hoon SK, Kim HC, Ku Y, Ro DH. Estimation of patient-reported outcome measures based on features of knee joint muscle co-activation in advanced knee osteoarthritis. Sci Rep. 2024 May 30;14(1):12428. doi:10.1038/s41598-024063266-7 (*first author)

40. Lee JY*, **Kim SE***, Kwon OH, Lee MC, Han HS, Ro DH. Shallow trochlear groove and narrow medial trochlear width at the proximal trochlea in patients with trochlear dysplasia: A three-dimensional computed tomography analysis. Knee Surg Sports Traumatol Arthrosc 2024;32(6):1434-1445. doi:10.1002/ksa.12166 (*first author)

41. **Kim SE**, Choi BS, Ro DH, Lee MC, Han HS. Fixed bearing and higher postoperative knee flexion angle as predictors of satisfaction in Asian patients undergoing posterior stabilized total knee arthroplasty. Clin Orthop Surg. 2024 Apr;16:e19. https://doi.org/10.4055/cios23166

42. **Kim SE**, Hwang S, Lee JH, Gimm G, Han HS, Choi BS, Ro DH. Minimal impact of beam projection angle deviations in skyline (Laurin) view and the efficacy of the anterior border of proximal tibia as a guiding landmark. Skelet. Radiol. 2024;53(7):1399-1406. doi:10.1007/s00256-024-04619-1

43. **Kim SE**, Han HS. Robotic arm-assisted total knee arthroplasty in a patient with osteopetrosis: a case report and review of literature. Orthop Rev (Pavia). 2024 Mar 1:16:94238. doi:10.52965/001c.94238

44. **Kim SE**, Nam JW, Kim JI, Kim JK, Ro DH. Enhanced deep learning model enables accurate alignment measurement across diverse institutional imaging protocols. Knee Surg Relat Res. 2024;36(1):4. doi:10.1186/s43019-023-00209-y

45. **Kim SE**, Kwak J, Ro DH, Lee MC, Han HS. The use of medial joint opening rather than mechanical axis deviation to determine the clinical outcomes after high tibial osteotomy. Bone Joint J 2024;106-B(3):240-248. doi:10.1302/0301-620X.106B3.BJJ-2023-0337.R3

46. **Kim SE**, Lee JH, Choi BS, Han HS, Lee MC, Ro DH. Performance of ChatGPT on Solving Orthopedic Board-Style Questions: a Comparative Analysis of ChatGPT 3.5 and ChatGPT 4. Clin Orthop Surg. 2024 Mar;16:e13. https://doi.org/10.4055/cios23179

47. **Kim SE**, Yun KR, Lee JM, Lee MC, Han HS. Preserving coronal knee alignment of the knee (CPAK) in unicompartmental knee arthroplasty correlates with superior patient-reported outcomes. Knee Surg Relat Res. 2024;36(1):1. doi:10.1186/s43019-023-00204-3.

48. **Kim SE**, Kim K, Lee MC, Han HS, Choi BS, Ro DH. An accurate method for measuring knee joint space width despite variations in beam projection angles. Knee. 2023;45:85-91. doi:10.1016/j.knee.2023.09.006

49. Choi BS, **Kim SE**, Yang M, Ro DH, Han HS. Functional alignment with robotic-arm assisted total knee arthroplasty demonstrated better patient-reported outcomes than mechanical alignment with manual total knee arthroplasty. Knee Surg Sports Traumatol Arthrosc. 2023;31(3):1072-1080. doi:10.1007/s00167-022-07227-5

50. **Kim SE**, Han HS. Robotic-assisted unicompartmental knee arthroplasty: historical perspectives and current innovations. Biomed Eng Lett. 2023;13(4):543-552. doi:10.1007/s13534-023-00323-6

51. **Kim SE**, Ro DH, Lee MC, Cholewa JM. Early- to Mid-Term Review of a Prospective, Multi-Center, International, Outcomes Study of an Anatomically Designed Implant with Posterior-Stabilized Bearing in Total Knee Arthroplasty. Medicina. 2023; 59(12):2105. https://doi.org/10.3390/medicina59122105

52. **Kim SE**, Lee MH, Cho CH, Lee J-I, Han H-S, Lee MC, Ro DH. Risk Factors and Clinical Outcomes of Osteotomy Plane Violation by D-Hole Screws in Medial Open Wedge High Tibial Osteotomy: A Simulation and Comparative Study. Medicina. 2023; 59(12):2104. https://doi.org/10.3390/medicina59122104

53. **Kim SE**, Jung H, Ro DH, Lee MC, Han HS. Anatomic Patellar Design Has No Clinical Superiority Over Dome Patellar Design in Total Knee Arthroplasty: A Randomized Clinical Trial. J Arthroplasty. 2023;38(9):1742-1747. doi:10.1016/j.arth.2023.03.056

54. **Kim SE**, Han HS, Lee MC, Ro DH. Single shot adductor canal block combined with intravenous patient-controlled analgesia can be effective as continuous adductor canal block in reducing opioid consumption and breakthrough pain after total knee arthroplasty. J Exp Orthop. 2022;9(1):84. Published 2022 Aug 23. doi:10.1186/s40634-022-00523-6

55. Park JY, **Kim SE**, Lee MC, Han HS. Elastic pneumatic tourniquet cuff can reduce postoperative thigh pain after total knee arthroplasty: a prospective randomized trial. BMC Musculoskelet Disord. 2020;21(1):565. Published 2020 Aug 21. doi:10.1186/s12891-020-03579-6

56. Rahim MK, **Kim SE**, So H, et al. Recent Trends in PET Image Interpretations Using Volumetric and Texture-based Quantification Methods in Nuclear Oncology. Nucl Med Mol Imaging. 2014;48(1):1-15. doi:10.1007/s13139-013-0260-2

Patents:

Kim HC, Ro DH, **Kim SE**. Method and Apparatus for Calculating Indicators Related to Knee Osteoarthritis by Using Electromyography Data. Republic of Korea Patent Application No. 10-2024-0007447. Published 2024 Jul 24. (Publication No. 10-2024-0114724).

Presentations:

Ryu B, Ro DH, Martin K, Inderhaug E, Persson A, Choi BS, Håland S, **Kim SE** Automated Posterior Tibial Slope Measurement Using Lateral Knee Radiographs: A Novel Landmark-Based Approach Using Deep Learning, 2025 Annual Meeting of the American Academy of Orthopaedic Surgeons, March 10, 2025, United States.

Kim SE, Ro DH, Lee MC, Han HS. Medial joint opening rather than mechanical axis deviation determined clinical outcomes after high tibial osteotomy, The 21st Congress of the European Society of Sports Traumatology, Knee Surgery and Arthroscopy (ESSKA), May 8–10, 2024, Milan, Italy.

Kim SE, MacDessi S, Song D, Kim JI, Choi BS, Han HS, Ro DH. Coronal Plane Alignment of the Knee (CPAK) Classification According to Kellgren and Lawrence grade, The 21st Congress of the European Society of Sports Traumatology, Knee Surgery and Arthroscopy (ESSKA), May 8–10, 2024, Milan, Italy.

Kim SE, Ro DH, Lee MC, Han HS. Automated posterior tibial slope measurement using deep learning algorithms in lateral knee radiographs, The 21st Congress of the European Society of Sports Traumatology, Knee Surgery and Arthroscopy (ESSKA), May 8–10, 2024, Milan, Italy.

Kim SE, Ro DH, Lee MC, Han HS. Does CPAK change in unicompartmental knee arthroplasty correlate with patient-reported outcome measures?, The 21st Congress of the European Society of Sports Traumatology, Knee Surgery and Arthroscopy (ESSKA), May 8–10, 2024, Milan, Italy.

Kim SE, MacDessi S, Song D, Kim JI, Choi BS, Han HS, Ro DH. Coronal Plane Alignment of the Knee (CPAK) Classification According to Kellgren and Lawrence grade, The 18th Annual Congress of Korean Society of Computer Assisted Orthopaedic Surgery. October 28, 2023, Republic of Korea

Kim SE, Kwak J, Ro DH, Lee MC, Han HS. Medial joint opening rather than mechanical axis deviation determined clinical outcomes after high tibial osteotomy, The 67th Annual Congress of Korean Orthopaedic Association, October 12-14, 2023, Republic of Korea.

Kim SE, Jung H, Ro DH, Lee MC, Han HS. Anatomic Patellar Design Has No Clinical Superiority Over Dome Patellar Design in Total Knee Arthroplasty: A Randomized Clinical Trial, The 67th Annual Congress of Korean Orthopaedic Association, October 12-14, 2023, Republic of Korea.

Kim SE, Lee MC, Ro DH, et al. Prospective Multi-center Outcomes Study of Persona Knee System in Total Knee Arthroplasty : Mid-term outcomes, The 66th Annual Congress of Korean Orthopaedic Association, October 13-15, 2022, Republic of Korea.

Kim SE, Hwang DH, Han HS, Lee MC, Ro DH, Low muscle mass is an independent risk factor for postoperative blood transfusion in total knee arthroplasty: A retrospective, propensity score-matched cohort study, The 40th Annual Congress of the Korean Knee Society, May 13-14, 2022, Republic of Korea.

Kim SE, Kwon SB, Jeon S, Kim M, Ro DH, Lee MC, Han HS, Relationship between Muscle Activation and Sagittal Knee Joint Biomechanics in Patients with Patellofemoral Pain Syndrome: a cross-sectional study, The 65th Annual Congress of Korean Orthopaedic Association, October 14-16, 2021, Republic of Korea.

Kim SE, Han HS, Lee MC, Ro DH, Pain control after TKA: Comparison between IV Patient controlled anesthesia, continuous adductor canal block, and single adductor canal block, The 64th Annual Congress of Korean Orthopaedic Association, October 15-17, 2020, Republic of Korea.

Kim SE, Ro DH, Kang T, Han DH, Lee DY, Han HS, Lee MC, Quantitative Evaluation of Gait features after Total Knee Arthroplasty: Comparison with Age and Sex-matched Controls, The 63rd Annual Congress of Korean Orthopaedic Association, October 17-19, 2019, Republic of Korea.

Luo L, Zhang X, **Kim SE**, et al., Assessment of Large Multimodal Reasoning Model Performance and Chain-of-Thought Quality in Chest Radiograph Interpretation, RSNA Cutting edge 2025

Professional Membership:

Member, The Korean Orthopaedic Association 2022-present

Member, The Korean Knee Society 2022-present

Member, The Korean Arthroscopy Society 2022-present

Member, The Korean Society of Computer Assisted Orthopaedic Surgery 2022-present

Member, American Academy of Orthopaedic Surgeons 2022-present

Member, Orthopaedic Research Society 2022-present

Awards:

Professional Awards

- **Highest Commendation from the Minister of Health and Welfare,** *Republic of Korea*, Apr 2017
    - Recognized for dedication and effective disease control measures as an Epidemic Intelligence Service Officer during the MERS crisis, significantly contributing to preventing the spread of the virus.

Academic Awards

- **Awarded Trainee Position**, *Harvard Medical School - Seoul National University Hospital – Seoul National University College of Medicine Collaborative Research Program*, Oct 2024
    - Selected for advanced AI development in knee MRI: Enhancing diagnostic accuracy and predictive capabilities.
    - Trainee: Sung Eun Kim, PI: Pranav Rajpurkar (Harvard Medical School).
    - Direct research funding: $250,000/year for 2 years (Total: $500,000 direct costs)
- **Best Paper Award**, *Knee Surgery & Related Research*, May 2024
    - Awarded for research on an enhanced deep learning model enabling accurate alignment measurement across diverse institutional imaging protocols.
- **Best Oral Presentation Award**, *The 18th Annual Congress of Korean Society of Computer-Assisted Orthopaedic Surgery*, Oct 2023.
- **Citation Award**, *Knee Surgery & Related Research*, May 2025
    - Awarded for publishing two highly cited articles in 2024, significantly contributing to the field of knee surgery and imaging research.
- **3rd Place**, *SNU Medicine Graduate Student Best Paper Award*, Aug 2023
    - Recognized for research: *Anatomic Patellar Design Has No Clinical Superiority Over Dome Patellar Design in Total Knee Arthroplasty: A Randomized Clinical Trial*.
- **Best Resident Award**, *Department of Orthopedic Surgery, Seoul National University Hospital*, Dec 2020
    - Honored for exemplary academic and clinical performance, inspiring fellow residents.
- **2010–2011: Superior Academic Performance Scholarship, Seoul National University**
- **2012: Merit-Based Scholarship, Seoul National University**
- **2012–2013: Scholarships Granted by Each College, Seoul National University**

Extracurricular Awards

- **Winner**, *National Medical Doctors Tennis Tournament*, Oct 2023
    - Represented Seoul National University College of Medicine Alumni. Acknowledged for outstanding sportsmanship and teamwork as co-captain, leading to victory.
- **Runner-Up**, *Korean Orthopedic Association Tennis Tournament*, Sep 2023
    - Achieved second place in a highly competitive event, demonstrating skill and dedication in the sport.
- **Grand Prize**, *The Seoul National University Hospital Group Slogan Contest*, Oct 2023
    - Created the slogan *"Beyond Excellence"* for the Seoul National University Hospital Group.

Editorial Activities:

Ad hoc Reviewer

- The Journal of Bone and Joint Surgery
- npj Digital Medicine

- Scientific Reports
- The Knee
- Knee Surgery & Related Research
- Clinics in Orthopedic Surgery
- BMC Musculoskeletal Disorders
- Archives of Orthopaedic and Trauma Surgery
- European Journal of Medical Research
- Journal of Orthopaedic Surgery and Research
- Arthroplasty
- Machine Learning for Health (ML4H)
- The Conference on Robot Learning (CoRL)